

Indian Statistical Institute

Course Name: M. Math, 2nd year

Subject Name : Topology III

Exam name : HW 3

Due: October 9, 2025

-
- You may use any results proved in class. Any other results (including those in homework problem sets) require proof. You are required to provide proper mathematical reasons for your answers.

1. For based path connected spaces X, Y , prove $\Sigma(X \times Y) \simeq \Sigma X \vee \Sigma Y \vee \Sigma(X \wedge Y)$.
2. Prove that for a fixed value of k , the sequences

$$\pi_k(U(1)) \rightarrow \pi_k(U(2)) \rightarrow \pi_k(U(3)) \rightarrow \pi_k(U(4)) \dots$$

$$\pi_k(O(1)) \rightarrow \pi_k(O(2)) \rightarrow \pi_k(O(3)) \rightarrow \pi_k(O(4)) \dots$$

eventually stabilize. Compute the groups $\pi_2 U(n)$, $\pi_2 SO(n)$ for all n .

3.
 - a) Prove that $\Omega CP^n \simeq S^1 \times \Omega S^{2n+1}$.
 - b) Suppose $S^r \rightarrow E \rightarrow X$ is a fibration such that the inclusion $S^r \rightarrow E$ is null-homotopic. Prove that $\Omega X \simeq S^r \times \Omega E$.